

Aditya Sugandhi

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SUMMARY

Software Engineer with 5+ years building **scalable data platforms** and **distributed systems**. Deep expertise in **data architecture**, **data modeling**, and **large-scale ETL pipelines** using **Python**, **SQL**, and **Java**. Proven track record designing **event-driven data orchestration** for **batch and streaming workloads** processing **50M+ daily events**. Strong technical leadership experience setting standards, owning systems end-to-end including **monitoring**, **reliability**, and **cost optimization**, and translating complex technical trade-offs to cross-functional stakeholders.

WORK EXPERIENCE

Software Engineer II

Florida State University — Finance Technology Division

Nov 2024 – Present

Tallahassee, FL

- Architected **distributed data platform** processing **5M+ daily telemetry events** from **35K+ IoT sensors** using **Apache Kafka** and **Spring Boot microservices**; designed **event-driven orchestration patterns** for **real-time streaming** and **batch workloads** achieving **99.99% availability**.
- Led **data architecture and modeling** initiatives for **time-series analytics platform**; implemented **horizontal PostgreSQL sharding** strategy for **500M+ records** improving query performance by **2x** and enabling **OLAP analytics** for compliance reporting.
- Built **large-scale ETL pipelines** using **Python** and **SQL** with **advanced performance optimization**; designed **distributed caching layer** with **Redis** reducing data synchronization latency by **45%** for real-time dashboards serving **10K+ concurrent users**.
- Owned systems **end-to-end including monitoring, reliability, and cost management**; established **observability infrastructure** with **Prometheus/Grafana**; implemented **automated alerting**, and optimized resource utilization reducing infrastructure costs by **30%**.
- Led **technical initiatives** across engineering teams; set **data platform standards** for schema design, data quality, and pipeline orchestration; mentored **6 engineers** on **distributed systems patterns** and **data engineering best practices**.
- Designed **37 RESTful API endpoints** with **real-time WebSocket streaming** for P&L calculations; translated **complex technical trade-offs** to finance stakeholders enabling data-driven decision making for compliance monitoring.

Machine Learning Engineer

Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024

Tallahassee, FL

- Built **scalable data pipelines** processing **1TB+ datasets** using **Python**, **Apache Spark**, and **SQL**; designed **batch and streaming orchestration** on **Kubernetes** achieving **50% pipeline runtime reduction** with **99% job success rate**.
- Implemented **data quality monitoring** and **observability dashboards** using **Grafana** and **MLflow**; established **automated data validation** and **anomaly detection** reducing data quality incidents by **70%**.
- Owned **end-to-end ML infrastructure** including **cost optimization**, **reliability monitoring**, and **performance tuning**; optimized compute resource allocation reducing cloud costs by **40%** while maintaining SLA targets.
- Collaborated with cross-functional teams translating **technical requirements** into **scalable data solutions**; delivered **data-driven insights** supporting strategic business decisions through **advanced SQL analytics** and reporting.

Software Engineer I

Aspire Systems

Oct 2020 – Jul 2022

Chennai, India

- Designed **data-intensive microservices** using **Java/Spring Boot** processing **500K+ events/day** with **Kafka streaming**; architected **event-driven data orchestration** for **financial reporting** achieving **99.5% availability**.
- Built **ETL pipelines** for **general ledger integration** and **P&L calculations** using **SQL** and **Python**; implemented **data modeling** for **regulatory compliance** and **audit trail** requirements supporting enterprise clients.
- Established **monitoring and reliability infrastructure** with **Prometheus/Grafana**; implemented **distributed tracing** and **automated alerting** reducing **MTTR** by **60%** and improving system observability.
- Led **technical initiatives** mentoring **5 junior engineers** on **data architecture patterns** and **performance optimization**; influenced **team standards** for code quality and **automated testing** improving velocity by **25%**.
- Developed **full-stack data applications** with **Angular/React** frontend and **RESTful APIs**; translated **technical trade-offs** to product stakeholders ensuring alignment on data platform roadmap.

Associate Software Engineer

IMPETUS Technologies

Jun 2019 – Sep 2020

Noida, India

- Developed **data processing services** using **Java** and **Python**; built **RESTful APIs** handling **100K+ daily requests** with **Redis caching** for low-latency data access supporting analytics dashboards.
- Implemented **workflow automation** and **data pipeline orchestration** using **Python** and **Node.js**; created internal tools improving engineering team productivity and data pipeline reliability.

PROJECTS

- **Enterprise Data Pipeline Orchestration** — Designed **Apache Airflow** workflow system processing **50M+ daily events** from financial data sources. Implemented **batch and streaming orchestration patterns**, **OLAP analytics**, and **automated monitoring**. Achieved **99.9% data accuracy** with **sub-second query latency**. Tech: **Python, SQL, Airflow, Spark, AWS**.
- **Real-Time Analytics Platform** — Architected **event-driven data platform** with **Java/Spring Boot** backend and **Angular** frontend. Built **streaming data pipelines** with **Kafka** and **PostgreSQL**. Implemented **distributed caching** and **monitoring infrastructure**. Tech: **Java, Python, Kafka, PostgreSQL, Redis**.
- **Data Quality Monitoring System** — Developed **automated data validation framework** using **Python** and **SQL**; built **observability dashboards** for **pipeline health monitoring** and **anomaly detection**. Integrated **alerting** reducing data quality incidents. Tech: **Python, SQL, Grafana, Prometheus**.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, Scala, TypeScript, JavaScript, HTML/CSS
- **Data Engineering:** Apache Kafka, Apache Spark, Apache Airflow, ETL/ELT Pipelines, Data Modeling, Data Architecture, Batch & Streaming Processing
- **Databases & Storage:** PostgreSQL (Sharding, Performance Tuning), Redis, MongoDB, Time-Series Databases, OLAP Systems
- **Cloud & Infrastructure:** AWS (EKS, Lambda, S3, EC2), Kubernetes, Docker, Distributed Systems, Microservices Architecture
- **Monitoring & Reliability:** Prometheus, Grafana, Distributed Tracing, Observability, Cost Optimization, SLA Management, Automated Alerting
- **Backend & APIs:** Spring Boot, Node.js, FastAPI, RESTful APIs, gRPC, Event-Driven Architecture, WebSocket Streaming
- **Frontend & Visualization:** Angular, React, Next.js, Power BI, Data Dashboards, TypeScript
- **Leadership & Practices:** Technical Leadership, Cross-functional Collaboration, Agile/Scrum, Code Quality Standards, Mentoring, Stakeholder Communication

EDUCATION

Florida State University M.S. in Computer Science	Tallahassee, FL
• Relevant Coursework: Distributed Systems, Advanced Database Systems, Machine Learning, Data Communications	
SRM Institute of Science & Technology B.Tech. in Computer Science	Chennai, India

PUBLICATIONS & CERTIFICATIONS

- Review Classification & False Feedback Detection (IJAST) | AWS Solutions Architect (Udemy, 2024)